# Module 09 Lab Reflective Journal

**John Grubbs**

I liked this assignment because it showed me how combining several models can make a machine learning system more reliable. The “wisdom of the crowd” idea helped me understand why a group of models can make better decisions than one model working alone.

I learned that a Random Forest combines many Decision Trees and has them vote on the final prediction. Using multiple trees can reduce overfitting because the model does not depend on the rules created by just one tree.

Both the single Decision Tree and the Random Forest reached 100% accuracy in this lab. This was expected because the Iris dataset is small, clean, and has clear differences between the flower species. The feature importance plot showed that petal width and petal length were the most important features, which makes sense because these measurements help distinguish the different types of iris flower.

**Keira Burdine**

In this lab, I learned how ensemble methods work and how combining multiple machine learning models can improve predictions. I worked with a single Decision Tree and a Random Forest using the Iris dataset. Both models achieved 100% accuracy, which surprised me because I expected the Random Forest to perform better.

One challenge I experienced was correcting the Python code formatting and troubleshooting errors. I had to identify missing information, correct the code and rerun cells until everything worked properly. This helped me become more comfortable with debugging and understanding how small errors can affect a program.

I also learned how to create a feature importance graph, which showed that petal width and petal length were the most important features for classifying the flowers. Overall, this lab helped me better understand ensemble learning, model accuracy and the importance of troubleshooting when working with machine learning.